Generalized Nonconvex Low-Rank Tensor Representation for Hyperspectral Anomaly Detection

Hao Qin, Qiangqiang Shen, Haijin Zeng, Yongyong Chen, *Member, IEEE*, and Guangming Lu, *Senior Member, IEEE*

***Abstract*— Low-rank tensor representation (LRTR) methods have attracted great interest for their powerful ability to separate backgrounds and anomalies. However, most of the current LRTR models use the popular and convex surrogate tensor nuclear norm (TNN) to solve optimization problems, which results in a loose approximation and suboptimal solver for the original problem. Besides, most existing methods solve the nonconvex optimization problems case-by-case, consequently losing one unified solver. To solve the above issues, we propose the generalized nonconvex low-rank tensor representation (GNLTR) for hyperspectral anomaly detection (HAD), a unified solver not case-by-case one of existing nonconvex optimization problems. Compared to the TNN, GNLTR contains many popular nonconvex penalty functions as tighter regularizers of the tensor tubal rank to constrain the low rank of the background. Moreover, the $L_{2,1}$ norm has been integrated into the GNLTR model for the sparse anomalies. For the optimization problem, it is handled quickly and efficiently through a well-organized alternating direction method of multipliers (ADMM). The experiments on several real-world hyperspectral datasets demonstrate the superior performance of the GNLTR model in comparison with some state-of-the-art anomaly detection (AD) models.**

***Index Terms*— Alternating direction method of multipliers (ADMM), hyperspectral anomaly detection (HAD), low-rank tensor representation (LRTR), unified anomaly detection (AD) nonconvex framework.**

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NOMENCLATURE

x, $\mathbf{x}$, $\mathbf{X}$, and $\mathcal{X}$	Scalar, vector, and matrix, tensor, respectively.
$\mathcal{X}^T$	Conjugate transpose of $\mathcal{X}$.
$\mathcal{I}$	Identity tensor.
$\mathcal{X}^{(i)}$	ith frontal slice of $\mathcal{X}$.
$\|\mathcal{X}\|_0$, $\|\mathcal{X}\|_1$, and $\|\mathcal{X}\|_{2,1}$	l_0, l_1, and $l_{2,1}$ norm of $\mathcal{X}$.
$\|\mathcal{X}\|_*$	Tensor nuclear norm of $\mathcal{X}$.
$\mathcal{A} * \mathcal{B}$	Tensor product of $\mathcal{A}$ and $\mathcal{B}$.
unfold($\mathcal{X}$)	Unfold $\mathcal{X}$ along mode-3.
rank$_a(\mathcal{X})$	Tensor average rank of $\mathcal{X}$.
rank$_t(\mathcal{X})$	Tensor tubal rank of $\mathcal{X}$.
$\bar{\mathcal{X}} = \text{fft}(\mathcal{X}, [], 3)$	Discrete Fourier transform of $\mathcal{X}$ along mode-3.

I. INTRODUCTION

IMAGING spectroscopy has greatly promoted the capture and processing of hyperspectral images (HSIs). Extensive hyperspectral imaging applications have emerged in recent years, such as military surveillance [1], [2], material identification [3], [4], grassland degradation monitoring [5], [6], invasive plant monitoring [7], [8], and image classification [9], [10]. Anomaly detection (AD) [11] occupies an important place in these applications since it could separate potentially valuable targets from HSIs. High-value targets considered anomalies, such as airplanes and vehicles, are usually embedded with a low probability in locally homogeneous backgrounds, which have different shapes, spectral features, and other characteristics from the surrounding pixels. As for the backgrounds such as mountains, rivers, and roads, their features are locally similar with a limited variation to each other. Therefore, AD could be regarded as a binary classification problem based on two primary principles: 1) anomalies appear with low probability in the background, while their spectral features differ from the surrounding background's spectral features and 2) the backgrounds are closely connected to each other due to the local similarity and limited variation from surrounding pixels.

Based on the above basic criteria, numerous methods have been formulated during the last decades. The pioneering work is the famous Reed–Xiaoli (RX) algorithm [12] as well as its variants, including global RX and local RX [13], dual-window-based RX [14], and multiple-window-based RX [15]. Nevertheless, the complexity and variability

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of hyperspectral data in practical applications usually cause difficulties in satisfying the multivariate normal distribution condition. To overcome this problem, various algorithms, such as subspace RX [16], weighted-RXD [17], and linear filter-based RXD [17], have been introduced. The weighted-RXD improved background estimation by altering the weights of background samples, anomalous pixels, or noisy signals, while the linear filter-based RXD filtered wrong anomalies or noise instances by using the probability. The kernel RX (KRX) [18] provided the kernelization procedure, which has the ability for mapping the raw data into high-dimensional space, such that the raw data can be separated linearly. The cluster KRX [19] classified pixels of backgrounds into different clusters with a fast eigendecomposition algorithm to detect anomalies.

In recent research, there is a tendency to focus on the inherent characteristics of HSIs. Prominent characteristics among them are the low rank of backgrounds and the anomalous scarcity. With this in mind, the robust principal component analysis (RPCA) framework [20] decomposed HSIs into a background part and sparse anomalies. The studies in [21] and [22] extended RPCA into the low-rank and sparse matrix decomposition (LRaSMD) in which the matrix was transformed into a low-rank matrix, a sparse matrix, and a noise matrix. The LRaSMD-based Mahalanobis distance (LSMAD) detector [23] took full advantage of the low rankness on the basis of LRaSMD. Niu et al. [24] enhanced the robustness and detection performance of their model with a learned dictionary and low-rank backgrounds. Xu et al. [25] proposed a sparsity-inducing regularized term into low-rank representation (LRR) with a new strategy to construct a comprehensive dictionary. The LRASMD-based spectral–spatial (LS-SS) method [26] integrated both spatial information and spectral information of the data into the model. Zhu et al. [27] focused on the sparse part and engaged a clustering-based weighting strategy. In [28], the graph regularization and total variation (TV) regularization were embedded into LRR (GTVLRR) for conservation of the spatial relationships and geometrical structure in HSIs. The $\mathbf{S}^2$ELR [29] method employed Schatten-p norm to approximate the rank and structured sparse norm, which plays an effective role in restricting the low-rank nature of the background and the sparsity of the anomalies.

However, the abovementioned methods usually transformed the 3-D HSI into a 2-D matrix, losing its intrinsic 3-D characteristics. In contrast, tensor [30], widely implemented in plenty of fields such as image fusion [31], [32], image restoration [33], [34], and subspace clustering [35], [36], is capable of preserving the inherent structure of multidimensional data. The first two dimensions of the 3-D HSIs stand for spatial information, while the third dimension is for spectral information. This drives researchers to conduct studies into tensor-based AD. Following the above research line, the low-rank tensor decomposition-based AD (LTDD) model [37] obtained a low-rank tensor with the Tucker decomposition plus a sparse tensor through an unmixing technology. In [38], tensor principal component analysis (TPCA) was carried out for gaining the principal component part and the residual part of HSIs. While prior-based tensor approximation (PTA) [39] had fully integrated a linear TV norm, a novel truncated nuclear norm (TNNR), and $L_{2,1}$ norm. Moreover, PCA-TLRSR [40] adopted a weighted tensor nuclear norm (TNN), a well-designed dictionary, and a principal component analysis (PCA) method for AD. In addition, many deep learning-based hyperspectral AD (HAD) methods also keep the 3-D characteristics of HSI. LREN [41] imported deep neural networks (DNNs) to construct an unsupervised low-rank embedded network with a specifically designed latent representation. DeCNN-AD [42] proposed a plug-and-play framework by integrating a convolutional neural network (CNN) denoiser and a new dictionary based on clustering. In PDBSNet [43] and B$\mathbf{S}^3$LNet [44], both of them proposed a novel blind-spot self-supervised reconstruction network, which rebuilt the center pixel's spectral signatures through neighbor pixels and considered the pixels that had with high reconstruction errors as anomalous pixels. For PDBSNet, it also used the pixel-shuffle downsampling method to break out the pixels' spatial relevance and enhanced the difference between anomalies and shuffled surrounding pixels.

Simultaneously, most low-rank-based models engaged the convex surrogate nuclear norm and its extensions as a rank substitute, but the effect is weaker than that of nonconvex surrogates [45]. Despite the wide exploration of tensor-based AD, the majority of these tensor-based approaches solve the nonconvex optimization problems case-by-case, lacking a unified framework in AD. For example, PTA [39] used the truncated nuclear norm, while PCA-TLRSR [40] adopted the weighted TNN as the nonconvex surrogates of matrix rank and tensor tubal rank. For solving their corresponding nonconvex optimization minimization problems, PTA [39] developed the two-step approach, while PCA-TLRSR [40] proposed the weighted tensor singular value thresholding. Consequently, a unified framework with nonconvex surrogates for AD should be put forward.

In this article, a unified nonconvex framework named generalized nonconvex low-rank tensor representation (GNLTR) is proposed for AD. The flow diagram of our GNLTR model is shown in Fig. 1. GNLTR transforms the HSI data into a dictionary derived from background information, the corresponding coefficient tensor, and an anomaly tensor. Our GNLTR provides a unified nonconvex surrogate in place of the familiar convex surrogate nuclear norm to restrict the low rankness of the backgrounds, allowing us to select the adaptive nonconvex penalty functions. For the anomaly tensor, GNLTR utilizes the $L_{2,1}$ norm to measure the dense property and the sparsity in the spectral direction. Finally, the alternating direction method of multipliers (ADMM) offers us an excellent option to solve the nonconvex optimization problem.

Compared with existing HAD models, the main contributions of this article are given as follows.

1) We propose the GNLTR framework for AD and a unified nonconvex surrogate function to approximate the tensor tubal rank, which fully takes advantage of the low-rank prior knowledge in AD.
2) A nonconvex, novel, and reliable dictionary construction approach is presented. Based on the reality that nonconvex surrogates are more approximate to rank_t compared with convex surrogates, we can better exploit the low-ranking property of the background part, which means that a purer background part could be separated from HSIs.

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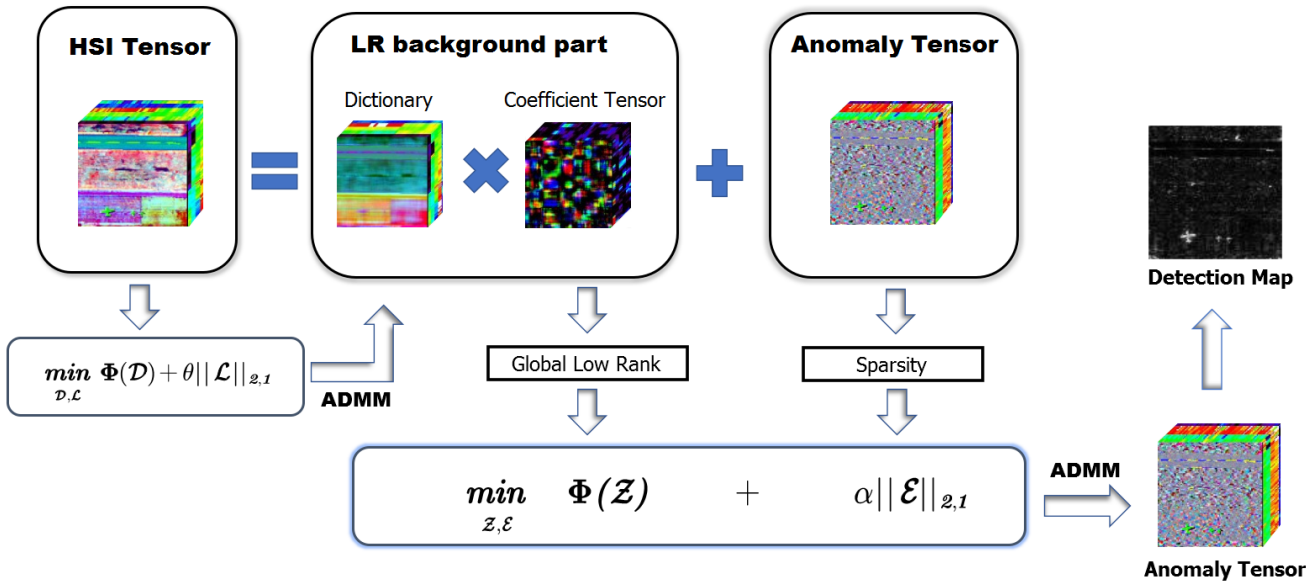


Fig. 1. Flow diagram of our proposed GNLTR model for HSI AD.

3) An iterative algorithm is derived based on the ADMM and its effectiveness is proven by extensive experiments on real-world HSIs.

The remainder of this article is organized as follows. Notations and preliminaries are presented in Section II. Section III introduces the unified AD nonconvex framework, the nonconvex dictionary construction, and their details. Section IV shows the experimental results and discussions. Conclusions are drawn in Section V.

II. Notations and Preliminaries

In this section, we introduce the notations mentioned in this article. For the sake of brevity, the main notations are presented in the Nomenclature. diag($\mathbf{x}$) is on behalf of diagonal matrix in which the diagonal is $\mathbf{x}$. diag($\mathbf{X}$) is a vector, which is consisted of $\mathbf{X}$'s diagonal elements. $\bar{\mathcal{X}}$ is denoted as the result of fast Fourier transform (FFT) along the 3th dimension of $\mathcal{X} \in \mathbb{R}^{n_1\times n_2\times n_3}$, i.e., $\bar{\mathcal{X}} = \text{fft}(\mathcal{X}, [], 3)$. Naturally, we could restore $\mathcal{X}$ with the inverse FFT, i.e., $\mathcal{X} = \text{ifft}(\bar{\mathcal{X}}, [], 3)$. Meanwhile, $\mathcal{X} \in \mathbb{R}^{n_1\times n_2\times n_3}$ could be transformed into a $n_1n_3 \times n_2$ matrix through the unfold operator, i.e., $\text{unfold}(\mathcal{X}) = [\mathcal{X}^{(1)}; \mathcal{X}^{(2)}; \ldots; \mathcal{X}^{(n_3)}]$. As for the block circulant matrix, it takes the slices of the tensor to construct an $n_3 \times n_3$ square matrix, which is defined as

$$\text{bcirc}(\mathcal{X}) = \begin{bmatrix} \mathcal{X}^{(1)} & \mathcal{X}^{(n_3)} & \mathcal{X}^{(n_3-1)} & \cdots & \mathcal{X}^{(2)} \\ \mathcal{X}^{(2)} & \mathcal{X}^{(1)} & \mathcal{X}^{(n_3)} & \cdots & \mathcal{X}^{(3)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \mathcal{X}^{(n_3)} & \mathcal{X}^{(n_3-1)} & \mathcal{X}^{(n_3-2)} & \cdots & \mathcal{X}^{(1)} \end{bmatrix}. \tag{1}$$

The following provides some definitions involved in this article.

Definition 1 (Tensor Product [46]): For two 3-D tensors $\mathcal{A} \in \mathbb{R}^{n_1\times n_2\times n_3}$ and $\mathcal{B} \in \mathbb{R}^{n_2\times n_4\times n_3}$, the tensor product $\mathcal{C} \in \mathbb{R}^{n_1\times n_4\times n_3}$ is

$$\mathcal{C} = \text{fold}(\text{bcirc}(\mathcal{A}) \cdot \text{unfold}(\mathcal{B})). \tag{2}$$

Definition 2 (Some Special Tensors): We list several definitions of special tensors:

1) *Conjugate Transpose Tensor [46]:* $\mathcal{A}^T \in \mathbb{R}^{n_2\times n_1\times n_3}$ denoted as the conjugate transpose of $\mathcal{A} \in \mathbb{R}^{n_1\times n_2\times n_3}$ is acquired by slicing along the third dimension with conjugate transposing each slice and then assembling each slice after reversing the order of previous arrangement.
2) *F-Diagonal Tensor [46]:* If each slice of a tensor $\mathcal{S}$ is a diagonal matrix, the tensor $\mathcal{S}$ is named f-diagonal.
3) *Identity Tensor [46]:* The identity tensor $\mathcal{I} \in \mathbb{R}^{n_1\times n_1\times n_2}$ is the one whose first frontal slice is an $n_1 \times n_1$ identity matrix and other frontal slices are zeros.
4) *Orthogonal Tensor [46]:* A tensor $\mathcal{Q} \in \mathbb{R}^{n_1\times n_1\times n_2}$ is orthogonal if it satisfies the equation $\mathcal{Q} * \mathcal{Q}^T = \mathcal{Q}^T * \mathcal{Q} = \mathcal{I}$.

Definition 3 (t-SVD [46], [47]): A tensor $\mathcal{X} \in \mathbb{R}^{n_1\times n_2\times n_3}$ has the factorization

$$\mathcal{X} = \mathcal{U} * \mathcal{S} * \mathcal{V}^T \tag{3}$$

where $\mathcal{S} \in \mathbb{R}^{n_1\times n_2\times n_3}$ should be a f-diagnoal tensor and $\mathcal{U} \in \mathbb{R}^{n_1\times n_1\times n_3}$ and $\mathcal{V} \in \mathbb{R}^{n_2\times n_2\times n_3}$ are orthogonal tensors.

Definition 4 (Tensor Tubal Rank [48]): For a tensor $\mathcal{X} \in \mathbb{R}^{n_1\times n_2\times n_3}$, its tensor tubal rank, denoted as $\text{rank}_t(\mathcal{X})$, is defined as the number of nonzero singular tubes of $\mathcal{S}$ with $\mathcal{X} = \mathcal{U} * \mathcal{S} * \mathcal{V}^T$. rank_t is

$$\text{rank}_t(\mathcal{X}) = \#\{i, \mathcal{S}(i, i, :) \neq \mathbf{0}\}. \tag{4}$$

Definition 5 (TNN [47]): First, we introduce the property induced by inverse DFT, formulated as

$$\mathcal{S}(i, i, 1) = \frac{1}{n_3}\sum_{j=1}^{n_3} \bar{\mathcal{S}}(i, i, j). \tag{5}$$

Second, $\mathcal{S}$ is f-diagonal, which means that $\#\{i, \mathcal{S}(i, i, :) \neq \mathbf{0}\}$ is equal to $\#\{i, \mathcal{S}(i, i, 1) \neq \mathbf{0}\}$. With these, the TNN can be defined as

$$\|\mathcal{X}\|_{\circledast} = \frac{1}{n_3}\sum_{i=1}^{m}\sum_{j=1}^{n_3} \bar{\mathcal{S}}(i, i, j) \tag{6}$$

where $m = \max(n_1, n_2)$.

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TABLE I
SOME POPULAR NONCONVEX PENALTY FUNCTIONS

Name	Formula	Supergradient	Name	Formula	Supergradient
Capped L_1 [52]	$\begin{cases}\lambda x, & x<\gamma,\\ \lambda\gamma, & x\geq\gamma.\end{cases}$	$\begin{cases}\lambda, & x<\gamma,\\ [0,\lambda], & x=\lambda,\\ 0, & x\geq\lambda.\end{cases}$	ETP [49]	$\frac{\lambda}{1-e^{-\gamma}}(1-e^{-\gamma x})$	$\frac{\lambda\gamma}{1-e^{-\gamma}}e^{-\gamma x}$
Geman [53]	$\frac{\lambda x}{\gamma+x}$	$\frac{\lambda\gamma}{(x+\gamma)^2}$	Laplace [54]	$\lambda(1-e^{-\frac{x}{\lambda}})$	$\frac{\lambda}{\gamma}e^{-\frac{x}{\gamma}}$
Logarithm [55]	$\frac{\lambda}{\log(\gamma+1)}\log(\gamma x+1)$	$\frac{\lambda\gamma}{(\gamma x+1)\log(\gamma+1)}$	L_p [50]	$\lambda x^p,\ 0<p<1$	$\begin{cases}\infty, & x=0,\\ \lambda p x^{p-1} & x>0.\end{cases}$
MCP [56]	$\begin{cases}\lambda x-\frac{x^2}{2\gamma}, & x\leq\gamma\lambda\\ \frac{1}{2}\gamma\lambda^2, & x>\gamma\lambda\end{cases}$	$\begin{cases}\lambda-\frac{x}{\gamma}, & x<\lambda\gamma,\\ 0, & x\geq\lambda\gamma.\end{cases}$	SCAD [51]	$\begin{cases}\lambda x, & x<\lambda,\\ \frac{-x^2+2\lambda\gamma x-\lambda^2}{2(\gamma-1)}, & \lambda\leq x<\gamma\lambda\\ \frac{\lambda^2(\gamma+1)}{2}, & x\geq\gamma\lambda\end{cases}$	$\begin{cases}\lambda, & x\leq\lambda,\\ \frac{\gamma\lambda-x}{\gamma-1}, & \lambda<x\leq\lambda\gamma,\\ 0, & x>\lambda\gamma\end{cases}$

Definition 6 (Tensor Average Rank [47]): The tensor average rank of a tensor $\mathcal{X}\in\mathbb{R}^{n_1\times n_2\times n_3}$, denoted as $\text{rank}_a(\mathcal{X})$, is defined as

$$\text{rank}_a(\mathcal{X})=\frac{1}{n_3}\text{rank}(\text{bcirc}(\mathcal{X})). \tag{7}$$

Definition 7 (Tensor Spectral Norm [47]): For a tensor $\mathcal{X}\in\mathbb{R}^{n_1\times n_2\times n_3}$, the tensor spectral norm is defined as

$$\|\mathcal{X}\|=\|\text{bcirc}(\mathcal{X})\| \tag{8}$$

where $\|\text{bcirc}(\mathcal{X})\|$ is the spectral norm for matrix $\text{bcirc}(\mathcal{X})$.

III. PROPOSED GNLTR MODEL AND THE NONCONVEX DICTIONARY CONSTRUCTION METHOD

In this section, a unified nonconvex surrogate of tensor tubal rank, a unified AD nonconvex framework, and a nonconvex dictionary construction method are introduced.

A. Unified Nonconvex Surrogate

Various kinds of nonconvex surrogates, such as the weighted TNN [40] and truncated nuclear norm [39] for specific nonconvex optimization problems, have been proposed in previous studies and have been broadly used in low-latitude spaces. Therefore, we integrate these nonconvex surrogates such as ETP [49], L_p [50], and SCAD [51] into a unified nonconvex surrogate structure, which applies different nonconvex penalty functions for different nonconvex optimization requirements.

Considering the structure of TNN, we can define the unified nonconvex surrogate for tensor tubal rank, i.e.,

$$\Phi(\mathcal{X})=\frac{1}{n_3}\sum_{i=1}^{m}\sum_{j=1}^{n_3}\phi(\bar{\mathcal{S}}(i,i,j)) \tag{9}$$

where $\mathcal{X}\in\mathbb{R}^{n_1\times n_2\times n_3}$, $m=\max(n_1,n_2)$, and $[\mathcal{U},\mathcal{S},\mathcal{V}]=\text{t-SVD}(\mathcal{X})$. Besides, some simple restrictions on the penalty function ϕ are needed here in order to satisfy the tensor tubal rank property. ϕ should be continuous, concave, and monotonically nondecreasing, which also meets

$$\begin{cases}\phi(0)=0\\ \lim\limits_{x\to\infty}\dfrac{\phi(x)}{x}=0.\end{cases} \tag{10}$$

What should be notable is that these requirements are simple and easy to meet. Based on the properties of ϕ, the inequality, i.e.,

$$\partial\phi(y)\leq\frac{\phi(x)-\phi(y)}{x-y}\leq\partial\phi(x) \tag{11}$$

holds true when $0<x<y$ and $x,y\in\mathbb{R}$. Table I summarizes some popular nonconvex penalty functions meeting the above restrictions. In these nonconvex penalty functions, λ is the scale parameter, γ is the shape parameter, and x is the independent variable. The above two variables λ and γ are greater than 0, with $x\geq 0$.

Moreover, we derive the properties of $\Phi(\mathcal{X})$ with these constraints of ϕ. $\Phi(\mathcal{X})$ is a continuous, monotonically nondecreasing, and concave function. As stated in Section I, we mentioned that the unified nonconvex surrogate with appropriate parameters is a tighter nonconvex envelope with respect to the TNN, and now, we provide the proof. On the set $\{\mathcal{X}\in\mathbb{R}^{n_1\times n_2\times n_3}|\ ||\mathcal{X}||\leq 1\}$, we can choose the appropriate parameters λ and γ with one parameter fixed and another parameter solved by the equation $\phi(1)_{\lambda,\gamma}=1$. Then, we hold $x\leq\phi_{\lambda,\gamma}(x)\leq 1$, which leads to

$$\|\mathcal{X}\|_{\circledast}\leq\Phi(\mathcal{X})\leq\text{rank}_a(\mathcal{X}). \tag{12}$$

In addition, the inequality, i.e., $\text{rank}_a(\mathcal{X})\leq\text{rank}_t(\mathcal{X})$, holds true for the tensor average rank and tensor tubal rank. Therefore, we claim that $\Phi(\mathcal{X})$ approximates tensor tubal rank better than $\|\mathcal{X}\|_{\circledast}$ with

$$\|\mathcal{X}\|_{\circledast}\leq\Phi(\mathcal{X})\leq\text{rank}_a(\mathcal{X})\leq\text{rank}_t(\mathcal{X}). \tag{13}$$

B. Unified AD Nonconvex Framework

1) Proposed GNLTR Model: Let HSI data be presented by a 3-D tensor $\mathcal{X}\in\mathbb{R}^{n_1\times n_2\times n_3}$, where n_3 indicates the amount of spectral bands and n_1 and n_2 show the spatial sizes of each frontal slice. We decompose $\mathcal{X}$ into backgrounds $\mathcal{L}\in\mathbb{R}^{n_1\times n_2\times n_3}$ and anomalies $\mathcal{E}\in\mathbb{R}^{n_1\times n_2\times n_3}$. The AD problem is replaced with

$$\begin{aligned}&\min_{\mathcal{L},\mathcal{E}}\text{rank}_t(\mathcal{L})+\alpha\|\mathcal{E}\|_{2,1}\\ &\text{s.t. }\mathcal{X}=\mathcal{L}+\mathcal{E}\end{aligned} \tag{14}$$

where α serves as a regularization parameter to balance the weight between backgrounds and anomalies, rank_t imposes restrictions on the low rank of backgrounds, and the $L_{2,1}$ norm

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Algorithm 1 Unified Nonconvex Surrogate Tensor Singular Value Thresholding

Input: Tensor $\mathcal{Z}^k$, Auxiliary variable $\mathcal{M}^k$, penalty parameter ρ^k, lagrangian multiplier β_1^k

Initialize: $[\mathcal{U}, \mathcal{S}^k, \mathcal{V}]$ =t-SVD($\mathcal{Z}^k$), $\bar{\mathcal{S}}^k = \text{fft}(\mathcal{S}^k, [], 3)$ $\mathcal{W} = \mathcal{M}^k - \frac{\beta_1^k}{\rho^k}$, $\bar{\mathcal{W}} = \text{fft}(\mathcal{W}, [], 3)$;

1: **for** i = 1 to $\lceil \frac{(n_3+1)}{2} \rceil$ **do**
2: $[\bar{\mathbf{U}}, \bar{\mathbf{S}}, \bar{\mathbf{V}}]$ =SVD $(\bar{\mathcal{W}}^{(i)})$;
3: $\mathbf{s}_d = diag(\bar{\mathbf{S}})$;
4: $\mathbf{s} = diag(\bar{\mathcal{S}}^{k(i)})$;
5: $\bar{\mathbf{S}}_* = diag(max(\mathbf{s}_d - \frac{\partial\phi(\mathbf{s})}{\rho^k}, 0))$;
6: $\bar{\mathcal{Z}}^{(i)} = \bar{\mathbf{U}} * \bar{\mathbf{S}}_* * \bar{\mathbf{V}}^T$
7: **end for**
8: **for** i = $\lceil \frac{(n_3+1)}{2} + 1 \rceil$ to n_3 **do**
9: $\bar{\mathcal{Z}}^{(i)} = conj(\bar{\mathcal{Z}}^{(n_3-i+2)})$;
10: **end for**
11: $\mathcal{Z} = \text{ifft}(\bar{\mathcal{Z}}, [], 3)$,

Output: $\mathcal{Z}$

is utilized for the spectral and spatial sparsity of anomalies. The $L_{2,1}$ norm is calculated by

$$\|\mathcal{E}\|_{2,1} = \sum_{i=1}^{n_1}\sum_{j=1}^{n_2}\sqrt{\sum_{t=1}^{n_3}\mathcal{E}(i,j,t)^2}. \tag{15}$$

However, calculating rank$_t$ is an NP-hard and nonconvex problem. To make the optimization problem computable, we take the unified nonconvex surrogate as a nonconvex alternative to rank$_t$, which could be reformulated as

$$\begin{aligned} &\min_{\mathcal{L},\mathcal{E}} \Phi(\mathcal{L}) + \alpha\|\mathcal{E}\|_{2,1} \\ &\text{s.t. } \mathcal{X} = \mathcal{L} + \mathcal{E} \end{aligned} \tag{16}$$

where $\Phi(\mathcal{L})$ is a unified nonconvex surrogate with ample nonconvex penalty functions.

Previous studies [23], [25], [40] have demonstrated that a reliable and well-designed dictionary has an important impact on improving the detection performance and robust representations of HSI data. Consequently, a comprehensive dictionary $\mathcal{D} \in \mathbb{R}^{n_1\times n_4\times n_3}$ extracted from the backgrounds without anomalies and a corresponding representation coefficient tensor $\mathcal{Z} \in \mathbb{R}^{n_4\times n_2\times n_3}$ imposed low-rank restrictions are inserted into the model to take the place of the background tensor. The renewed framework is

$$\begin{aligned} &\min_{\mathcal{Z},\mathcal{E}} \Phi(\mathcal{Z}) + \alpha\|\mathcal{E}\|_{2,1} \\ &\text{s.t. } \mathcal{X} = \mathcal{D} * \mathcal{Z} + \mathcal{E}. \end{aligned} \tag{17}$$

2) Optimization Solver: To figure out the optimization problem, ADMM offers a clear, straightforward, and efficient method. First, the tensor $\mathcal{Z}$ is excepted to be decoupled from the restriction $\mathcal{X} = \mathcal{D} * \mathcal{Z} + \mathcal{E}$ to reduce the complexity of the following optimization formulas. We introduced an auxiliary variable $\mathcal{M}$ to replace $\mathcal{Z}$, and (17) is rewritten as

$$\begin{aligned} &\min_{\mathcal{Z},\mathcal{M},\mathcal{E}} \Phi(\mathcal{Z}) + \alpha\|\mathcal{E}\|_{2,1} \\ &\text{s.t. } \mathcal{Z} = \mathcal{M}, \quad \mathcal{X} = \mathcal{D} * \mathcal{M} + \mathcal{E}. \end{aligned} \tag{18}$$

Then, we transform (18) into the augmented Lagrangian function, i.e.,

$$\begin{aligned} \min_{\mathcal{Z},\mathcal{M},\mathcal{E}} \Phi(\mathcal{Z}) + \alpha\|\mathcal{E}\|_{2,1} + \langle\beta_1, \mathcal{Z} - \mathcal{M}\rangle + \frac{\rho}{2}\|\mathcal{Z} - \mathcal{M}\|_F^2 \\ + \langle\beta_2, \mathcal{X} - \mathcal{D} * \mathcal{M} - \mathcal{E}\rangle + \frac{\rho}{2}\|\mathcal{X} - \mathcal{D} * \mathcal{M} - \mathcal{E}\|_F^2 \end{aligned} \tag{19}$$

where ρ works as a penalty parameter and β_1 and β_2 are Lagrangian multipliers. Equation (19) is an unconstrained problem with six unknown variables. By fixing other unknowns, we update one unknown each time. The following are the optimization equations for each unknown variable.

1) *Update $\mathcal{Z}$:* After extracting the items related to $\mathcal{Z}$, the formula is

$$\mathcal{Z}^{k+1} = \arg\min_{\mathcal{Z}} \Phi(\mathcal{Z}) + \frac{\rho^k}{2}\left\|\mathcal{Z} - \mathcal{M}^k + \frac{\beta_1^k}{\rho^k}\right\|_F^2 \tag{20}$$

where $\mathcal{Z}^{k+1}$ is the result of the $(k+1)$th iteration of $\mathcal{Z}$ and the same for others. Equation (20) can be transformed to

$$\begin{aligned} \mathcal{Z}^{k+1} = \arg\min_{\mathcal{Z}} \sum_{j=1}^{n_3}\left(\sum_{i=1}^{m}\phi(\bar{\mathcal{S}}(i,i,j))\right) \\ + \frac{\rho^k}{2}\|\mathcal{Z}^{(j)} - \mathcal{H}^{(j)}\|_F^2 \end{aligned} \tag{21}$$

where $\mathcal{H} = \mathcal{M}^k - (\beta_1^k/\rho^k)$, $\mathcal{Z} \in \mathbb{R}^{n_1\times n_2\times n_3}$, $m = \min(n_1, n_2)$, $\mathcal{Z} = \mathcal{U} * \mathcal{S} * \mathcal{V}^T$, and $\mathcal{Z}^{(j)}$ stands for the jth frontal slice of $\mathcal{Z}$. Moreover, we can derive the inequality, i.e.,

$$\begin{aligned} \phi(\bar{\mathcal{S}}(i,i,j)) \le \phi(\bar{\mathcal{S}}^k(i,i,j)) - \partial\phi(\bar{\mathcal{S}}^k(i,i,j)) \\ \times \left(\bar{\mathcal{S}}^k(i,i,j) - \bar{\mathcal{S}}(i,i,j)\right) \end{aligned} \tag{22}$$

through (11). With (22), the optimization problem (21) can be relaxed to

$$\begin{aligned} \mathcal{Z}^{k+1} = \arg\min_{\mathcal{Z}} \sum_{j=1}^{n_3}\left(\sum_{i=1}^{m}\partial\phi(\bar{\mathcal{S}}^k(i,i,j))\bar{\mathcal{S}}(i,i,j)\right) \\ + \frac{\rho^k}{2}\|\mathcal{Z}^{(j)} - \mathcal{H}^{(j)}\|_F^2. \end{aligned} \tag{23}$$

It is obvious that (23) is composed of separated n_3 problems and the optimal solution could be obtained by solving the subproblems. The jth subproblem is

$$\begin{aligned} \mathcal{Z}^{k+1(j)} = \arg\min_{\mathcal{Z}^{(j)}}\left(\sum_{i=1}^{m}\partial\phi(\bar{\mathcal{S}}^k(i,i,j))\bar{\mathcal{S}}(i,i,j)\right) \\ + \frac{\rho^k}{2}\|\mathcal{Z}^{(j)} - \mathcal{H}^{(j)}\|_F^2. \end{aligned} \tag{24}$$

The optimal solution of the above equation (24) can be obtained with the generalized weighted singular value thresholding (GWSVT) [57], [58], [59], i.e.,

$$\bar{\mathcal{Z}}^{k+1(j)} = \bar{\mathbf{U}} * \bar{\mathbf{S}}_* * \bar{\mathbf{V}}^T \tag{25}$$

where $[\bar{\mathbf{U}}, \bar{\mathbf{S}}, \bar{\mathbf{V}}] = \text{SVD}(\bar{\mathcal{H}}^{(j)})$, $\mathbf{s}_d = \text{diag}(\bar{\mathbf{S}})$, $\mathbf{s} = \text{diag}(\bar{\mathcal{S}}^{k(j)})$, $\bar{\mathbf{S}}_* = \text{diag}(\max(\mathbf{s}_d - ((\partial\phi(\mathbf{s}))/\rho^k), 0))$, and $\mathcal{Z}^{k+1} = \text{ifft}(\bar{\mathcal{Z}}^{k+1}, [], 3)$. The whole process of updating $\mathcal{Z}$ is shown in Algorithm 1.

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2) *Update $\mathcal{E}$:* It is updated by

$$\mathcal{E}^{k+1} = \arg\min_{\mathcal{E}} \alpha\|\mathcal{E}\|_{2,1} + \frac{\rho^k}{2}\|\mathcal{E} - \mathcal{P}\|_F^2 \tag{26}$$

where $\mathcal{P} = \mathcal{X} - \mathcal{D} * \mathcal{M}^k + (\beta_2^k/\rho^k)$. For the subproblem, its closed-form solution is

$$\mathcal{E}(:,:,t) = \begin{cases} \dfrac{\|\mathcal{P}(:,:,t)\|_2 - \frac{\alpha}{\rho^k}}{\|\mathcal{P}(:,:,t)\|_2}\mathcal{P}(:,:,t), \\ \qquad \dfrac{\alpha}{\rho^k} < \|\mathcal{P}(:,:,t)\|_2 \\ 0, \quad \text{otherwise.} \end{cases} \tag{27}$$

3) *Update $\mathcal{M}$:* The function with respect to $\mathcal{M}$ is

$$\mathcal{M}^{k+1} = \arg\min_{\mathcal{M}} \frac{\rho^k}{2}\left\|\mathcal{Z}^{k+1} - \mathcal{M} + \frac{\beta_1^k}{\rho^k}\right\|_F^2 + \frac{\rho^k}{2}\left\|\mathcal{X} - \mathcal{D} * \mathcal{M} - \mathcal{E}^{k+1} + \frac{\beta_2^k}{\rho^k}\right\|_F^2. \tag{28}$$

The optimization problem comprises two squares of tensor F-norm. We set the differential coefficient of (28) to be $\mathbf{0}$, i.e.,

$$(\mathcal{D}^T * \mathcal{D} + \mathcal{I})\mathcal{M} - \left(\mathcal{Z}^{k+1} + \frac{\beta_1^k}{\rho^k} + \mathcal{D}^T * \left(\mathcal{X} - \mathcal{E}^{k+1} + \frac{\beta_2^k}{\rho^k}\right)\right) = \mathbf{0}. \tag{29}$$

Equation (29) is linear, which indicates that $\mathcal{M}$ achieves a minimum at the point where the differential coefficient is $\mathbf{0}$, i.e.,

$$\mathcal{M}^{k+1} = (\mathcal{D}^T * \mathcal{D} + \mathcal{I})^{-1} * \left(\mathcal{Z}^{k+1} + \frac{\beta_1^k}{\rho^k} + \mathcal{D}^T * \left(\mathcal{X} - \mathcal{E}^{k+1} + \frac{\beta_2^k}{\rho^k}\right)\right). \tag{30}$$

4) *Update β_1, β_2, and ρ:* These variables are updated by

$$\beta_1^{k+1} = \beta_1^k + \rho^k(\mathcal{Z}^{k+1} - \mathcal{M}^{k+1}) \tag{31}$$

$$\beta_2^{k+1} = \beta_2^k + \rho^k(\mathcal{X} - \mathcal{D} * \mathcal{M}^{k+1} - \mathcal{E}^{k+1}) \tag{32}$$

$$\rho^{k+1} = \max(r\rho^k, \rho_{\max}) \tag{33}$$

where the growth rate r is usually set to 1.1. It has been demonstrated that the update strategy of ρ can accelerate the ADMM convergence.

In [60], the theoretical proof of ADMM convergence with more than two block variables is tough. Therefore, we could consider ADMM as converged when the changes in all variables are less than a small value $\varepsilon = e^{-6}$. The convergence condition is

$$\max\begin{pmatrix} \|\mathcal{Z}^{k+1} - \mathcal{Z}^k\|_\infty, & \|\mathcal{E}^{k+1} - \mathcal{E}^k\|_\infty \\ \|\mathcal{M}^{k+1} - \mathcal{M}^k\|_\infty, & \|\mathcal{M}^{k+1} - \mathcal{Z}^{k+1}\|_\infty \\ \|\mathcal{X} - \mathcal{D} * \mathcal{M}^{k+1} - \mathcal{E}^k\|_\infty & \end{pmatrix} \le \varepsilon. \tag{34}$$

Finally, we can get the AD map $\mathbf{T}$ by

$$\mathbf{T}(i,j) = \sqrt{\sum_{t=1}^{n_3} |\mathcal{E}(i,j,t)|^2} \tag{35}$$

where $\mathcal{E} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ and $\mathbf{T} \in \mathbb{R}^{n_1 \times n_2}$. We render all the procedures in Algorithm 2.

Algorithm 2 GNLTR for HAD

Input: HSI data $\mathcal{X}$, dictionary $\mathcal{D}$, regularization parameter α
Initialize: $\mathcal{Z}^0, \mathcal{M}^0, \mathcal{E}^0, \beta_1^0, \beta_2^0$ initialized to $\mathbf{0}$; $\mu = 1.1$, $\rho^0 = e^{-5}$, $\rho_{max} = e^8$, $\varepsilon = e^{-6}$, $max_iter = 100$, $k = 0$;
while not converged and $k \le max_iter$ **do**
  Update $\mathcal{Z}^{k+1}$ via Algorithm 1;
  Update $\mathcal{E}^{k+1}$ via Eq. (27);
  Update $\mathcal{M}^{k+1}$ via Eq. (30);
  Update $\beta_1^{k+1}, \beta_2^{k+1}, \rho^{k+1}$ via Eqs. (31), (32), (33), respectively;
  Check the converge condition Eq. (34);
  $k = k + 1$;
end while
Obtain the optimal coefficient tensor $\mathcal{Z}^*$ and anomalies $\mathcal{E}^*$;
Compute the backgrounds $\mathcal{L}$ via $\mathcal{L} = \mathcal{D} * \mathcal{Z}^*$;
Compute the detection map $\mathbf{T}$ by Eq. (35);
Output: backgrounds $\mathcal{L}$, anomalies $\mathcal{E}^*$ and detection map $\mathbf{T}$.

C. Dictionary Construction

In Section III-B, a comprehensive dictionary $\mathcal{D}$ and a corresponding representation coefficient tensor $\mathcal{Z}$ are utilized for representing the backgrounds. The dictionary should consist of pure background parts so as to obtain a better representation. Here, we provide a nonconvex, feasible, and effective dictionary construction technique that can effectively separate backgrounds and anomalies, i.e.,

$$\min_{\mathcal{D},\mathcal{L}} \Phi(\mathcal{D}) + \theta\|\mathcal{L}\|_{2,1} \quad \text{s.t. } \mathcal{X} = \mathcal{D} + \mathcal{L} \tag{36}$$

where $\mathcal{L}$ is the anomalous part. To solve this optimization problem, we also exploit ADMM. The augmented Lagrangian function transformed by (36) is

$$\min_{\mathcal{D},\mathcal{L}} \Phi(D) + \theta\|\mathcal{L}\|_{2,1} + <\beta, \mathcal{X} - \mathcal{D} - \mathcal{L}> + \frac{\rho}{2}\|\mathcal{X} - \mathcal{D} - \mathcal{L}\|_F^2 \tag{37}$$

where $\Phi(\mathcal{D})$ is the nonconvex alternative to $\text{rank}_t(\mathcal{D})$, β is a Lagrangian multiplier, and ρ is a penalty parameter. ADMM is a good choice to address the optimization [see (37)]. The subproblems are similar to (19), which means that we could use Algorithm 2 to handle the subproblems with a few modifications. The details are given as follows.

1) *Update $\mathcal{D}$:* The dictionary is updated by

$$\mathcal{D}^{k+1} = \arg\min_{\mathcal{D}} \Phi(\mathcal{D}) + \frac{\rho^k}{2}\left\|\mathcal{X} - \mathcal{D} - \mathcal{L}^k + \frac{\beta^k}{\rho^k}\right\|_F^2. \tag{38}$$

The equation has a similar structure to (20), which can be solved by Algorithm 1.

2) *Update $\mathcal{L}$:* The function related to $\mathcal{L}$ is

$$\mathcal{L}^{k+1} = \arg\min_{\mathcal{L}} \theta\|\mathcal{L}\|_{2,1} + \frac{\rho^k}{2}\|\mathcal{L} - \mathcal{P}\|_F^2 \tag{39}$$

where $\mathcal{P} = \mathcal{X} - \mathcal{D}^{k+1} + (\beta_k/\rho_k)$. We have noted that (39) has the same composition as (26), and thus, we can

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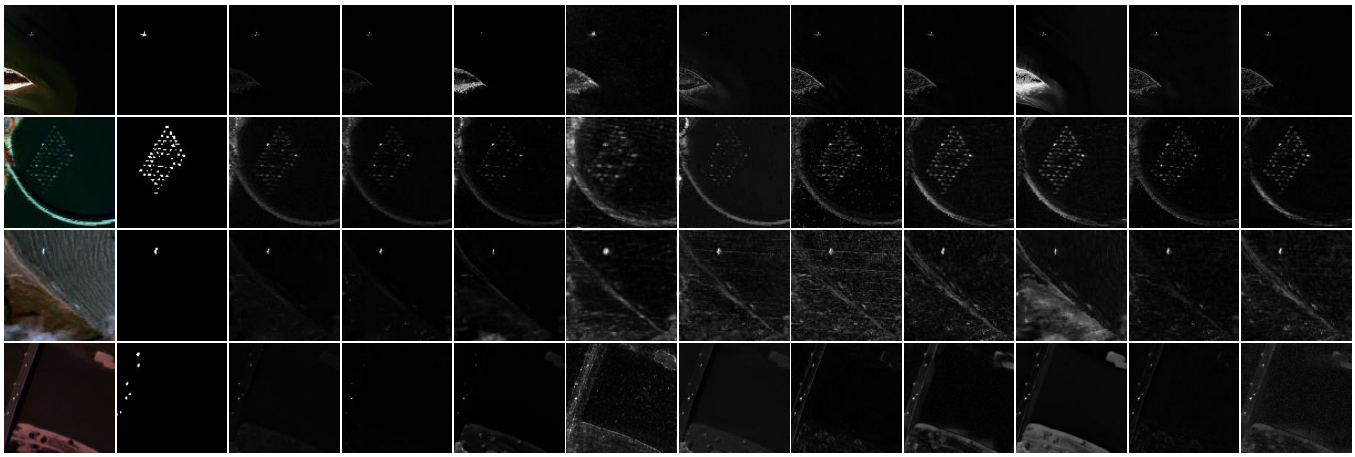


Fig. 2. Pseudo-color images, reference maps, and detection maps on the Beach data. The top-to-bottom images are Beach-1–Beach-4. (a) HSIs. (b) Reference. (c) RX. (d) RPCA. (e) LSMAD. (f) TPCA. (g) LRSAR. (h) GTVLRR. (i) TLRSR. (j) LREN. (k) DeCNN. (l) GNLTR.

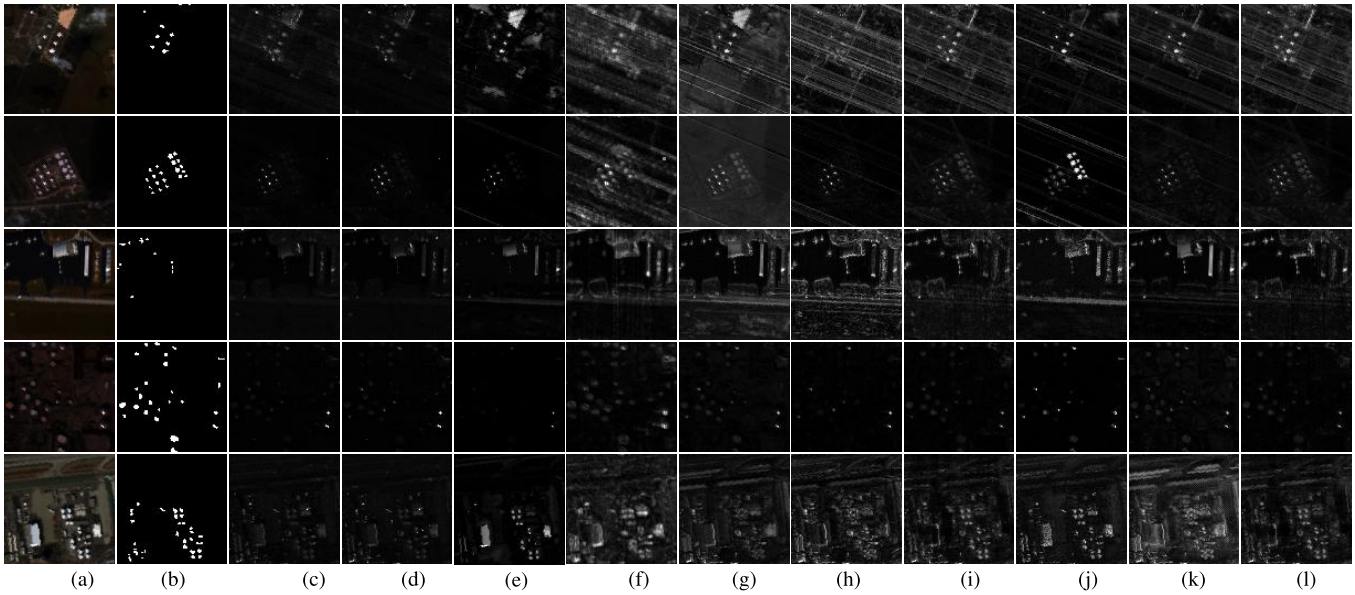


Fig. 3. Pseudo-color images, reference maps, and detection maps on the Urban data. The top-to-bottom images are Urban-1–Urban-5. (a) HSIs. (b) Reference. (c) RX. (d) RPCA. (e) LSMAD. (f) TPCA. (g) LRSAR. (h) GTVLRR. (i) TLRSR. (j) LREN. (k) DeCNN. (l) GNLTR.

obtain a closed-form solution by replacing the symbols of (27).

3) *Update β and ρ:* The two variables could be updated by

$$\beta^{k+1} = \beta^k + \rho^k(\mathcal{X} - \mathcal{D}^{k+1} - \mathcal{L}^{k+1}) \tag{40}$$

$$\rho^{k+1} = \max(r\rho^k, \rho_{\max}). \tag{41}$$

Notably, the dictionary $\mathcal{D}$ is a 3-D tensor that preserves the intrinsic spatial–spectral architecture of the HSI data $\mathcal{X}$. Also, the $L_{2,1}$ norm plays an important role in eliminating the impact of anomalies and the nonconvex surrogate $\Phi(\cdot)$ brings a purer dictionary than the convex surrogate due to the closer approximation to rank$_t$.

IV. Experimental Results

In this section, we introduce several real-world hyperspectral datasets to assess the detection performance and effectiveness of our proposed GNLTR model. All the experimental results were obtained by a computer with an Intel Core i5-10400 2.90-GHz CPU plus 16-GB memory.

A. Hyperspectral Datasets Description

The testing datasets include Beach, Urban, HYDICE-Urban, and San Diego. Some details about those datasets are summarized as follows.

1) In the Beach dataset, ships, vehicles, flights, and man-made objects are labeled as anomalies. Meanwhile, beaches and water are considered as backgrounds. Two images of the data have an image size of 100 × 100. The other two images are 150 × 150. Fig. 2(a) shows the pseudo-color images. Fig. 2(b) shows reference detection maps.
2) The Urban dataset contains five 100 × 100 images derived from AVIRIS. Roofs and other objects that are significantly different from their surroundings will be viewed as anomalies. We present the detailed images in Fig. 3(a) and (b).

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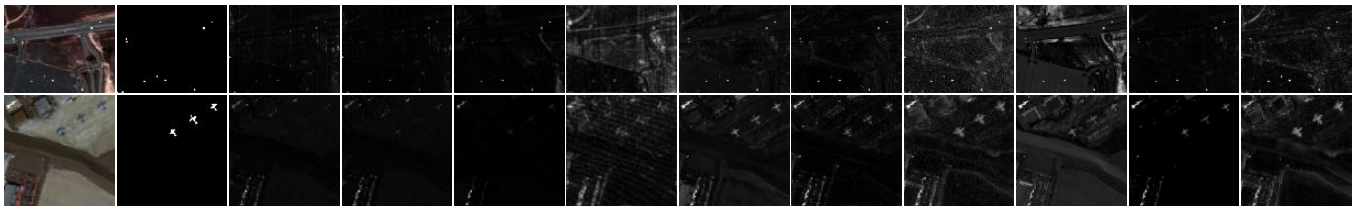


Fig. 4. Pseudo-color images, reference maps, and detection maps on (top) HYDICE-urban and (bottom) San Diego. (a) HSIs. (b) Reference. (c) RX. (d) RPCA. (e) LSMAD. (f) TPCA. (g) LRSAR. (h) GTVLRR. (i) TLRSR. (j) LREN. (k) DeCNN. (l) GNLTR.

TABLE II
PERFORMANCES OF DIFFERENT NONCONVEX PENALTY FUNCTIONS FOR DICTIONARY CONSTRUCTION AND L_p FOR RESULT SOLVING ON AUC ON SEVERAL DATASETS

Functions	Capped L_1	ETP	Geman	Laplace	Logarithm	L_p	MCP	SCAD
Beach-2	0.9753	0.9622	0.9640	0.9622	0.9609	0.9648	0.9638	0.9584
Urban-3	0.9918	0.9919	0.9872	0.9879	0.9907	0.9914	0.992	0.9922
HYDICE-Urban	0.9951	0.9956	0.9949	0.9962	0.9948	0.9963	0.9961	0.9948
San Diego	0.9954	0.9956	0.9940	0.9949	0.9955	0.9957	0.9957	0.9954

3) The HYDICE-Urban with $80 \times 100 \times 175$ pixels is urban scenes originated from the Hyperspectral Digital Imagery Collection Experiment (HYDICE). Roofs and vehicles are considered anomalies, which contain 21 pixels in the different parts of the detection map. Fig. 4(a) and (b) shows the sample images and related reference maps.
4) San Diego is with $100 \times 100 \times 189$ pixels. Backgrounds are mostly made of normal things such as grass and roofs. As to the anomalies, three airplanes are naturally marked as the main components of the exceptions with details given in Fig. 4(a) and (b).

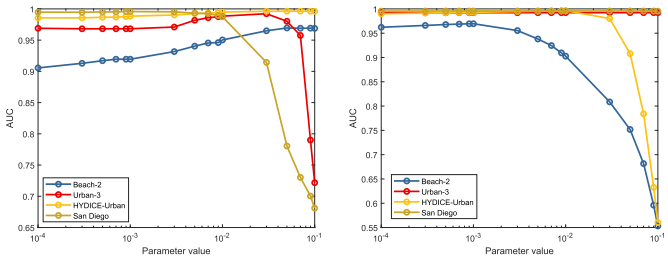


Fig. 5. Hyperparameters tuning on four datasets. (a) θ for dictionary construction. (b) α for result solving.

B. Parameter Tuning

For the proposed GNLTR model, four variables are supposed to be optimized to gain satisfactory detection results. The first two hyperparameters θ and α involved in (17) and (36) are selected in $\{0.0005, 0.001, 0.01, 0.03, 0.05, 0.07, 0.1\}$. The other two are in connection with the nonconvex surrogate selection, which is chosen in Table I. One of these parameters is adjusted while others are fixed. Fig. 5 shows the robustness of the proposed model for θ and α, which means that GNLTR is insensitive to the parameters over a relatively large range on most datasets. Table II gives the performance of different nonconvex penalty functions used to build dictionary and L_p for result solving on area under the receiver operating characteristics (AUCs) with appropriate parameters on some datasets. Meanwhile, Table III shows the performances of different nonconvex penalty functions for result solving and L_p for dictionary construction. Notably, the AUCs of different nonconvex penalty functions have little variation, which implies that our unified nonconvex framework is well structured with excellent performance.

C. Detection Performance

To examine the results, some baselines are compared with the proposed GNLTR, which include classic RX, four LRaSMD models (i.e., RPCA [20], LSMAD [23], LRASR [25], and GTVLRR [28]), two tensor representation models (i.e., TPCA [38] and PCA-TLRSR [40]), and two deep learning-based HAD methods (i.e., LREN [41] and DeCNN-AD [42]). In addition, we optimize the parameters in LSMAD, LARSR, and GTVLRR to achieve first-class detection accuracy. For LSMAD, the parameter rank varies from 1 to 30. In terms of LARSR, two hyperparameters λ and β are selected from $\{0.001, 0.01, 0.05, 0.1, 0.5, 1, 2\}$. As to GTVLRR, there are three regularization parameters with the set point, i.e., $\beta = 0.2$, $\gamma = 0.05$, and λ varies from 0.05 to 1. For PCA-TLRSR, we get optimal values for two parameters according to the suggestion in [40]. Considering the parameters used in [41], we set the hidden nodes $= 9$, $k = 7$, and λ from 10^{-4} to 1 for LREN. With regard to DeCNN-AD, K varies from 1 to 20, and the values of β and λ change from 10^{-4} to 1. Moreover, the number of clusters is set to 15 and each cluster as atoms for the dictionary owns 20 pixels in LRASR and GTVLRR. As for our model, the parameters are programmed to the suitable values, which has been discussed in parameter tuning. In addition, we adopt PCA to preprocess the original data to improve the effectiveness and reduce the computation time.

To qualitatively and quantitatively assess the detection performance, we provide visual observation and three evaluation indicators of the results. Before introducing the specific metrics, we briefly explain the meaning of P_d, P_f, and τ, which stand for probability of detection, false alarm rate (FAR), and threshold value, respectively. The first indicator is the receiver operating characteristic (ROC) curve in Figs. 6 and 7. The ROC of (P_d, P_f) depicts the varying correlation between P_d and P_f. A better model is considered to own a higher P_d

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TABLE III

PERFORMANCES OF DIFFERENT NONCONVEX PENALTY FUNCTIONS FOR RESULT SOLVING AND L_p FOR DICTIONARY CONSTRUCTION ON AUC IN SEVERAL DATASETS

Functions	Capped L_1	ETP	Geman	Laplace	Logarithm	L_p	MCP	SCAD
Beach-2	0.9626	0.9629	0.9695	0.9695	0.9658	0.9648	0.9631	0.9626
Urban-3	0.9907	0.9912	0.9911	0.9911	0.9913	0.9914	0.9913	0.9913
HYDICE-Urban	0.9960	0.9961	0.9959	0.9959	0.9963	0.9963	0.9958	0.9960
San Diego	0.9957	0.9957	0.9958	0.9958	0.9958	0.9957	0.9957	0.9957

TABLE IV

AUC VALUES OF (P_d, P_f) ON ALL TESTING DATASETS

HSI data	RX	RPCA	LSMAD	TPCA	LRASR	GTVLRR	PCA-TLRSR	LREN	DeCNN-AD	GNLTR
Beach-1	0.9804	0.9761	0.8808	**0.9890**	0.9260	0.9773	0.9875	0.9115	0.9737	0.9880
Beach-2	0.9106	0.9097	0.9033	0.8061	0.8812	0.9306	0.9638	0.9039	0.9411	**0.9695**
Beach-3	**0.9999**	0.9995	0.9296	0.9982	0.9987	0.9954	0.9993	0.9011	0.9995	0.9993
Beach-4	0.9539	0.9599	0.8959	0.9583	0.9552	**0.9815**	0.9643	0.8919	0.9739	0.9795
Average	0.9612	0.9613	0.9024	0.9379	0.9403	0.9712	0.9787	0.9021	0.9721	**0.9841**
Urban-1	0.9907	0.9922	0.9612	0.9390	0.9685	0.8465	**0.9949**	0.9786	0.9914	0.9947
Urban-2	0.9946	0.9957	0.9783	0.9409	0.9974	0.8995	**0.9983**	0.9981	0.9638	0.9979
Urban-3	0.9513	0.9577	0.9697	0.8224	0.9336	0.9328	0.9919	0.9209	0.9799	**0.9923**
Urban-4	0.9887	0.9871	0.9902	0.9617	**0.9932**	0.9373	0.9884	0.9905	0.9916	0.9887
Urban-5	0.9692	0.9658	0.9115	0.9370	0.9115	0.9391	**0.9837**	0.9554	0.9571	0.9830
Average	0.9789	0.9797	0.9622	0.9202	0.9608	0.9110	**0.9914**	0.9687	0.9678	0.9913
HYDICE-Urban	0.9857	0.9842	0.8818	0.8242	0.9926	0.9746	0.9941	0.9539	**0.9976**	0.9963
San Diego	0.8886	0.9165	0.9893	0.8849	0.9920	0.9869	0.9957	0.9892	0.9913	**0.9958**

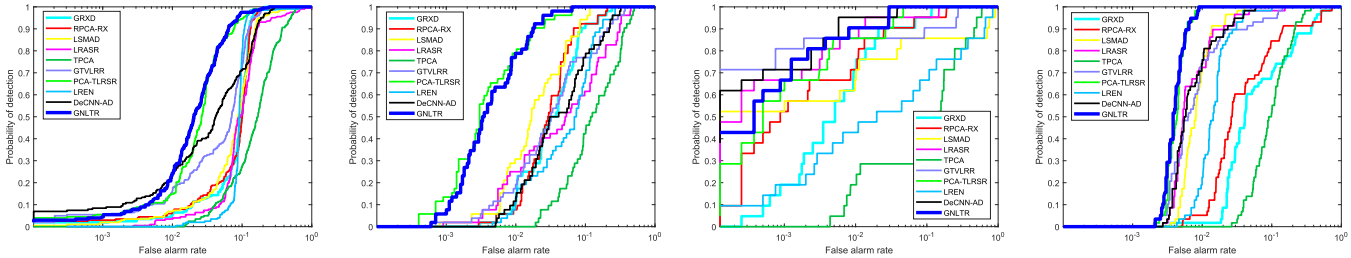


Fig. 6. ROC curves of (P_d, P_f) on different datasets (left to right) Beach-2, Urban-3, HYDICE-Urban, and San Diego.

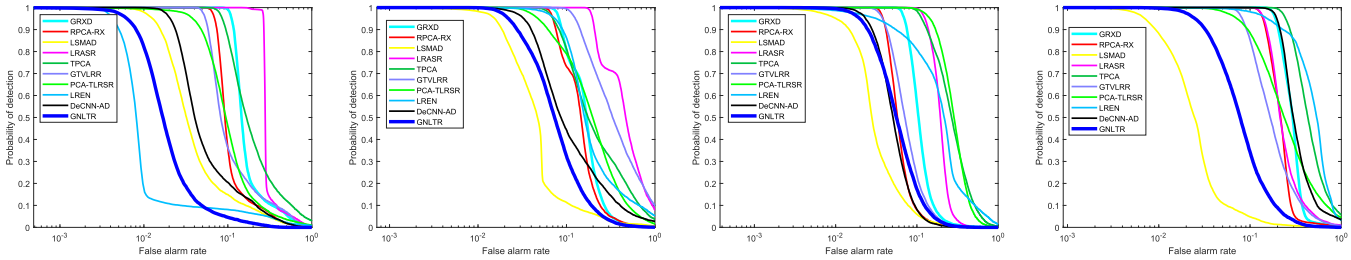


Fig. 7. ROC curves of (P_f, τ) on different datasets (left to right) Beach-2, Urban-3, HYDICE-Urban, and San Diego.

for a given P_f. The relations between P_f and τ can be represented by the ROC of (P_f, τ). A lower P_f indicates a lower likelihood that a method misclassifies a pixel for a given τ. The second one is the AUC in Tables IV and V. For AUC of (P_d, P_f), the closer the value of the method is to 1, the more accurate the method's detection is. In contrast, the lower AUC of (P_f, τ) means that the method has a lower probability of false detection in general. The third metric is the separability boxplot, which counts the detection results at the pixel locations of the anomaly and background categories with a statistical range from 10% to 90%. A more separable method owns a larger gap between the blue background box and the red anomaly box. With these evaluation indicators, the detection performance is given as follows.

1) *Beach:* Four scenes of Beach are shown in Fig. 2 with the correlative AUC scores in Tables IV and V. Here, we employ Beach-2 to demonstrate the outstanding result of GNLTR. The first three methods and LRASR rarely detect the anomalies. TPCA is poor in maintaining the shape of anomalies. GTVLRR, PCA-TLRSR, LREN, DeCNN-AD, and GNLTR successfully remain anomalies' forms. However, GTVLRR and DeCNN-AD do not fully detect the anomalies. At the same time, GNLTR views the bay more as a background than an anomaly, which is better aligned with the detection map. In terms of the Beach-2's ROC of (P_d, P_f) which is shown in Fig. 6(a), we find that DeCNN-AD gets the first place at the beginning and GNLTR is in the lead with PCA-TLRSR in the second place when $P_f > 0.01$. In ROC of (P_f, τ) of Beach-2 presented in Fig. 7(a), GNLTR gains the lowest value after $P_f > 0.05$. For AUCs of (P_d, P_f), GNLTR is the best on Beach-2

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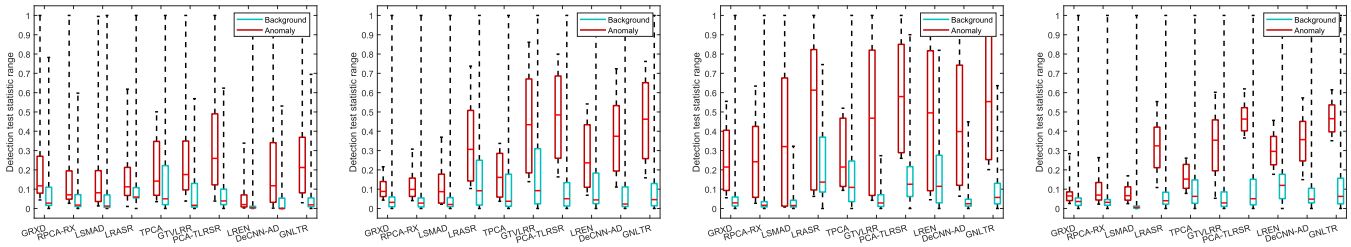


Fig. 8. Separability maps for different datasets (left to right) Beach-2, Urban-3, HYDICE-Urban, and San Diego.

TABLE V
AUC VALUES OF (P_f, τ) ON ALL TESTING DATASETS

HSI data	RX	RPCA	LSMAD	TPCA	LRASR	GTVLRR	PCA-TLRSR	LREN	DeCNN-AD	GNLTR
Beach-1	0.0113	**0.0096**	0.0248	0.0491	0.0819	0.0331	0.0175	0.0159	0.0323	0.0225
Beach-2	0.0803	0.0569	0.0312	0.1079	0.1367	0.0577	0.0569	0.0192	0.0353	**0.0119**
Beach-3	0.0473	0.0292	**0.0254**	0.0857	0.1273	0.1187	0.0701	0.1155	0.0445	0.0906
Beach-4	0.0296	**0.0065**	0.0137	0.0968	0.0714	0.0283	0.0544	0.0706	0.0246	0.0440
Average	0.0422	0.0244	**0.0238**	0.0849	0.1043	0.0594	0.0497	0.0553	0.0342	0.0423
Urban-1	0.0932	**0.0549**	**0.0549**	0.1733	0.2543	0.1838	0.1334	0.0588	0.0782	0.1183
Urban-2	0.0193	0.0188	**0.0054**	0.1190	0.1357	0.0280	0.0434	0.0258	0.0477	0.0156
Urban-3	0.0501	0.0454	0.0312	0.0838	0.1590	0.1546	0.0715	0.0771	0.0521	**0.0274**
Urban-4	0.0185	0.0181	**0.0005**	0.0399	0.0319	0.0114	0.0168	0.0045	0.0314	0.0032
Urban-5	0.0734	0.0610	**0.0328**	0.1678	0.1242	0.1239	0.0882	0.0883	0.1948	0.0382
Average	0.0509	0.0396	**0.0250**	0.1168	0.1410	0.1003	0.0707	0.0509	0.0808	0.0405
HYDICE-Urban	0.0615	0.0349	**0.0220**	0.1610	0.1052	0.0442	0.1617	0.1347	0.0309	0.0364
San Diego	0.0670	0.0510	**0.0087**	0.1128	0.0629	0.0543	0.0748	0.1206	0.0828	0.0263

and the second on Beach-1 and Beach-4. Moreover, for AUCs of (P_f, τ), GNLTR obtains the first position on Beach-2. As a result, GNLTR has a noteworthy improvement in the average scores on (P_d, P_f) and a quality result on the scores of (P_f, τ). In the separability map of Beach-2 [Fig. 8(a)], GNLTR has the largest gap with narrow widths. In addition, the median of backgrounds is close to 0, which indicates that our method has a strong ability to suppress backgrounds.

2) *Urban:* Five urban scenarios make up these data, which are observed in Fig. 3. In these scenes, Urban-3 is a pretty good option as a representative of Urban for its typical backgrounds and anomalies. RX, RPCA, LSMAD, and TPCA do not own the capability to fully detect anomalies. The maps of LRSAR, GTVLRR, LREN, and DeCNN-AD make mistakes in treating a great deal of the background as anomalies. PCA-TLRSR and GNLTR successfully suppress the backgrounds better while detecting anomalies. For the relevant ROCs of (P_d, P_f) shown in Fig. 6(b), PCA-TLRSR gets the first place when $P_f < 0.007$. GNLTR overtakes PCA-TLRSR and keeps the first when $P_f > 0.007$. GNLTR is only behind LSMAD in ROCs of (P_f, τ) in Fig. 7(b). Besides, the related AUCs of (P_d, P_f) are available in Table IV. GNLTR occupies the first position on Urban-3 followed by PCA-TLRSR and the second place on Urban-1. GNLTR obtains the suboptimal effect in the average scores, whose value is only 0.01 less than PCA-TLRSR. For AUC values of (P_f, τ) in Table V, GNLTR is the first on Urban-3 and second on Urban-2 and Urban-3. The average value is also excellent. In Fig. 8(b), the map shows that PCA-TLRSR has the largest gap, tightly followed by GNLTR. What is more, the two boxes of the GNLTR have a smaller width relative to the PCA-TLRSR.

3) *HYDICE-Urban:* Fig. 4 shows us a view of the detection results. Unfortunately, RX, RPCA, and TPCA ignore the anomalies. The remaining methods show the ability to detect anomalies. But DeCNN-AD and GNLTR perform better among all compared models because of the clearest anomalies' forms and a relatively clean background. In Figs. 6(c) and 7(c) and Tables IV and V, we provide the ROC curves and AUC values. Although, for ROCs of (P_d, P_f), GNLTR is at the middle place when $P_f < 0.012$, it gains the highest P_d in the following. In regard to ROCs of (P_f, τ), GNLTR gets the second place when $P_f < 0.05$ and the fourth in the remaining part. For AUCs of (P_d, P_f), GNLTR leaves all comparison methods far behind with the second highest value of 0.9963 except for DeCNN-AD. The GNLTR's score of (P_f, τ) obtains the fourth. In Fig. 8(c), our model has the biggest gap and there is no overlap between the two boxes.

4) *San Diego:* Three high-value targets are distributed at the top-right part of the image and can be visually observed in Fig. 4. It is obvious that RX, RPCA, LSMAD, and TPCA fail to discover the anomalies. At the same time, the maps of LRASR, GTVLRR, LREN, and DeCNN-AD lose the clear form of planes. On the contrary, PCA-TLRSR and GNLTR acquire a satisfactory structure. Considering the ROC curves in Fig. 6(d), PCA-TLRSR and GNLTR have a larger area than other compared methods. The P_d values of GNLTR are tightly followed by PCA-TLRSR. P_d of GNLTR achieves 1 at FAR $= 0.007$. In Fig. 7(d), GNLTR is the second one overall. Moreover, the AUCs of (P_d, P_f) in Table IV indicate that GNLTR has the highest values, tightly followed by PCA-TLRSR. The AUCs of (P_f, τ) in Table V show that the GNLTR is the second lowest value. For the separability map in Fig. 8(d),

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TABLE VI
TIME CONSUMPTION (S) OF GNLTR AND BASELINES ON SOME DATASETS

HSIs	RX	RPCA	LSMAD	TPCA	LRASR	GTVLRR	PCA-TLRSR	LREN	DeCNN-AD	GNLTR
Beach-2	0.068	2.608	8.581	16.280	14.612	108.887	7.381	190.996	62.021	4.003
Urban-3	0.075	2.786	8.275	17.245	31.040	122.447	6.483	228.201	28.130	6.321
HYDICE-Urban	0.047	1.948	5.801	10.930	22.055	102.337	6.038	148.185	150.800	2.264
San Diego	0.062	2.532	8.414	15.741	27.980	116.145	2.253	183.336	30.590	2.087

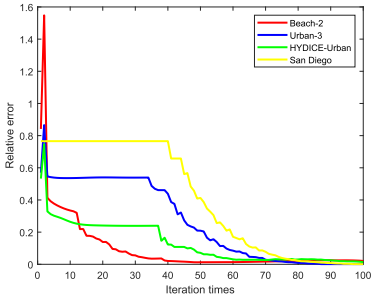


Fig. 9. Relative error for different iteration times on four datasets.

the proposed method owns the biggest interval and the highest anomaly median.

D. Discussion

1) Computational Cost: Table VI shows the running time of GNLTR and all compared methods on some datasets. Due to the superior architecture, GNLTR's time consumption is excellently controlled, which has a lower cost than LSMAD, TPCA, LRASR, GTVLRR, PCA-TLRSR, LREN, and DeCNN-AD. As to the computational complexity of Algorithm 2, there are five parts. First, updating $\mathcal{Z}$ via Algorithm 1 takes about $O(n_2^3 n_3)$. For $\mathcal{M}$ and $\mathcal{E}$, they cost $O(n_1 n_2^3 n_3 + n_2(n_1 + n_3)n_3 \log(n_3))$. With regard to β_1, β_2, and ρ, they could be finished in $O(n_1 n_2^2 n_3)$ time. In summary, the total complexity is $O((n_1+1)n_2^3 n_3 + n_2(n_1+n_2)n_3 \log(n_3) + n_1 n_2^2 n_3)$.

2) Convergence Analysis: In Fig. 9, we provide the numerical convergence results on four datasets in which the relative errors are the max part of (34). For Beach-2, Urban-3, and HsYDICE-Urban, their relative errors hit the maximum value at times $= 1$ and then monotonically decline. Although the error of San Diego keeps stable at the beginning, it decreases after times $= 40$. It is obvious that all the errors finally approach 0 when times >70.

V. CONCLUSION

In this article, we have proposed a novel unified AD nonconvex framework for HAD. Tensor has been integrated into the framework to preserve the intrinsic 3-D property of HSI. A unified nonconvex surrogate in the model has a better approximation than the TNN in terms of the tensor tubal rank, which means that it denotes the low rankness of the backgrounds more effectively. The $L_{2,1}$ norm regularization is an excellent choice to limit the sparsity of the anomaly component. ADMM is employed as the optimization solver. In addition, we extract a nonconvex dictionary construction method for AD. Experiments have proven the outstanding performance of our model in contrast with other existing AD methods. For future work, the linear TV-norm regularization can be adopted to denote the piecewise-smooth characteristic of backgrounds.

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